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Perspective

Environmental pollution as a critical driver of antimicrobial resistance emergence

Sunidhi Bhatt[1] · **Subhankar Chatterjee**[2]

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Abstract
Environmental processes are becoming known as paramount determinants of the emergence and spread of antimicrobial resistance (AMR). Antimicrobial residues, resistant microbes, and co-selective pollutants can be found in various environmental matrices at levels that drive the development of resistance. The sub-inhibitory concentrations of antimicrobials, heavy metals, microplastics, and biocides contribute to genetic selection as well as horizontal transfer of resistance determinants, which facilitate the development of persistent environmental reservoirs. Through such exposures, resistance determinants are selected, allowing for an interaction between environmental reservoirs and human and animal interfaces. This evidence notwithstanding, the majority of AMR frameworks still pay significantly more attention to clinical and agricultural misuse, and the significant environmental pathways are not adequately covered. This perspective article highlights the need to integrate environmental protection into the central part of the One Health response to AMR. Pharmaceutical industries, hospitals and municipal wastewater production, animal rearing, and domestic releases fill ecosystems with biologically active compounds, which increase in resistance and mobility. Additionally, the hydrological variation caused by climate can redistribute contaminants, thereby expanding the geographical area of contamination hotspots associated with AMR. A key set of priorities revealed through this article is to improve the environmental aspect of AMR governance, such as establishing enforceable discharge levels for antibiotic-contaminated waste, developing treatment technologies that can potentially remove co-selective pollutants, enhancing monitoring of environments considered at risk, and aligning responsibility among contributing sectors. Such considerations should be incorporated into AMR policies in countries and globally to mitigate the environmental impact of drug resistance and safeguard the future effectiveness of antimicrobial agents.

Keywords Antimicrobial resistance · Antibiotics · Selective pressure · One health · Environment

AMR is a worldwide crisis whose dimensions have long been due for attention, but one aspect has recently received overdue attention: the environment. Research indicates that the presence of antibiotic-resistant bacteria and resistance genes is common, not just in hospitals, but also in rivers, soils, sediments, and even the air. Natural ecosystems have been silently transforming into reservoirs and amplifiers of AMR. It is in opposition to the traditional AMR strategies, which emphasize human medicine and animal medicine exclusively. Misuse could be brought under control by hospital wards

✉ Sunidhi Bhatt, bhattsunidhi0@gmail.com; ✉ Subhankar Chatterjee, schatt.pondiuni@gmail.com; drsubhankar.eco@pondiuni.ac.in | [1]Bioremediation and Metabolomics Research Group, Department of Environmental Sciences, Central University of Himachal Pradesh, Academic Block Shahpur, District Kangra, City Shahpur, Himachal Pradesh 176206, India. [2]Bioremediation and Metabolomics Research Group, Department of Ecology & Environmental Sciences, School of Life Sciences, Pondicherry University, R.V. Nagar, Kalapet, City Pondicherry, Puducherry 605014, India.

and well-managed farms. Still, all their efforts may go to waste when massive amounts of active antibiotics remain free to enter the environment. To maintain the effective use of drugs, experts caution that we should not overlook the environmental aspect of AMR and people should position it as part of the One Health agenda, rather than a peripheral matter. If AMR "cuts across human, animal health, and environmental concerns" [1], why have major national action plans left antibiotics flowing freely into nature? O'Neill's review and related policy discussions focused on improved stewardship and new drug development, but concrete measures to limit environmental emissions have been minimal. For instance, a country might strictly control on-farm antibiotic sales yet have no requirement for treating waste from farms or hospitals, even when it contains active drugs. Such gaps make the fight against AMR incomplete, creating loopholes through which AMR re-enters communities from the environment.

Many recent global models have synchronized with the scale of this blind spot. Based on modelled data representing approximately one-third of total human antibiotic consumption, a recent study estimates that roughly 8,500 tonnes of antibiotics are introduced into the world's rivers annually [2]. Even with the operational sewage treatment, the secondary treatment of sewage effluents results in the discharge of large quantities of antimicrobials and a significant population of resistant microbiota into the aquatic environment [3]. It has been shown that sub-inhibitory concentrations of these pollutants impose selective pressure on the indigenous microbial communities, driving the proliferation of AMR determinants [4]. The concentrations of antibiotics present in untreated municipal sewage are usually lower than those in hospital effluents. Still, they may be sufficiently high to have a selective pressure, particularly under the influence of contributions from clinical facilities or pharmaceutical effluents [5]. Extending this analysis to the entire contaminant portfolio, the estimated annual release of over 8000 tonnes excludes contributions from intensive livestock production and specialized pharmaceutical manufacturing facilities, whose direct environmental inputs are known to be substantially elevated [6]. These observations suggest that anthropogenic activities have created large, persistent antibiotic-enriched ecosystems that function effectively as selective bioreactors, accelerating the emergence and dissemination of AMR determinants on a continental scale. All sectors of society contribute to this pollution. Hospitals, clinics, and residences flush unmetabolized drugs into sewage systems, while pharmaceutical manufacturing sites often discharge highly concentrated antibiotic effluent into nearby rivers. Animals fed antibiotics emit a significant percentage of the given dose in unmetabolized form, which subsequently leach into the surrounding soil and waterways through manure and wastewater. Aquaculture adds to the problem: fish farms dosing their stock with antibiotics release medicated water directly into coastal ecosystems. Daily domestic routines, including bathing, laundry, and waste disposal, can also contribute to sewage systems with trace levels of pharmaceuticals; however, it is unclear how these inputs contribute to the environmental resistome, and it may be worthwhile to explore this subject in greater detail [7].

Even the most carefully managed health systems are not spared from the flow-through of waste antibiotics into the environment via the WWT pathways; environmental microbes exposed to such waste remain vulnerable to developing antibiotic resistance. Considering concrete examples of broad-spectrum fluoroquinolone antibiotics, which are vital in both medicine and veterinary care, epitomizes the challenge. We previously reported that approximately 70% of administered fluoroquinolones are excreted unchanged, with major contributions from pharmaceutical effluent, hospital waste, and agricultural runoff [8]. Farm animals contribute, too: medicated feed and therapy result in antibiotics in manure, which is then spread on fields. This persistent presence in the environment generates intense selection pressure, driving the emergence of multidrug-resistant bacteria. The very potency of these drugs in hospitals and barns becomes their liability once they spill back into nature. These inputs often coincide with other pollutants, such as heavy metals, biocides, or even microplastics in the environment, which can synergize with antibiotics to select for multidrug resistance. In fact, a new study found that adding diverse microplastic particles to soil significantly increased the abundance of antibiotic resistance genes (ARGs) in microbial communities [9]. In short, ecosystems become hotspots for the evolution of AMR wherever antibiotics co-occur with other stressors, such as metals, plastics, or chemical waste.

The ecological effects extend beyond microbes. Antibiotic pollution can disrupt vital ecosystem services. For instance, antibiotics in agricultural soils inhibit nitrogen-fixing bacteria, altering plant–microbe symbioses and potentially affecting crop growth and soil fertility. In aquatic environments, residues can impair microbial food webs and disturb nutrient cycling. Moreover, these environmental pools of AMR do not respect borders or species lines. Antimicrobial-resistant microbes can migrate from wildlife reservoirs or contaminated water bodies into human populations through food chains or physical contact. A systematic review indicated that reused water for agriculture, frequently deriving from municipal or agricultural waste, can introduce antibiotics and select for resistant bacteria on produce [10]. At the same time, seafood harvested from contaminated coastal areas may convey AMR determinants acquired from adjacent waters.

Airborne transmission is another potential emerging route of AMR. It has been documented that aerosols resulting from animal husbandry, wastewater treatment, and soils can contain antibiotic residues as well as antibiotic-resistant

bacteria and genes [11]. Recognized hotspots of airborne AMR include concentrated animal feeding operations and sewage treatment plants [11]. Recent data from urban settings established that pollution is a significant factor associated with increased concentrations of antibiotic resistance genes (ARGs) in the atmosphere [12]. Therefore, antibiotic-resistant microbes and genes present in bioaerosols are capable of migrating far from their sources, and thus may facilitate the dispersal of antibiotic resistance to remote ecosystems and unsuspecting populations. Such pathways erect a loop whereby environmental exposure re-incorporates AMR into the human microbiome, partially circumventing institutional advances in hygiene and antimicrobial stewardship.

Climate fluctuations also exacerbate the pollution-triggered mechanisms of AMR. Projected increases in air and water temperatures, alongside heightened frequency and intensity of extreme meteorological events, seem to accelerate the propagation of AMR in environmental compartments. For instance, increasing atmospheric temperatures expedite bacterial proliferation and horizontal gene transfer, with each 10 °C elevation correlating with a 4–5% increase in AMR rates among significant pathogens, notably *Escherichia coli* and *Klebsiella pneumoniae*, measured against a panel of relevant antibiotics [13, 14]. Similarly, hydrological extremes disseminate drug contaminants and antibiotic-resistant microorganisms throughout landscapes. Flooding events can discharge antibiotic-laden wastewater and resistant microorganisms into streams, soils, and aquifers, while droughts concentrate these contaminants in diminishing reservoirs, increasing exposure [15]. Consequently, climatic extremes interact synergistically with chemical pollutants to augment concealed environmental reservoirs of AMR, thereby disseminating resistance genes into previously unexposed regions and populations [15]. Thus, ecological disruption by climate change and pollution can act synergistically: climate shifts amplify every hidden reservoir of AMR, expanding AMR risk into new regions and communities. Recognizing these dangers, global health leaders have begun to take action; however, current efforts fall short of meeting the challenge.

In late 2024, the WHO released its first guidance for antibiotic manufacturing waste, bluntly binding regulations [16]. Yet, as of now, this is largely a recommendation. Additionally, worldwide One Health efforts now specifically mention environmental AMR. The WHO's "Tricycle" surveillance program provides a standardized protocol for monitoring ESBL-producing *E. coli* across humans, animals, and environmental samples [17]. UNEP's 2023 report, "Bracing for Superbugs," emphasizes that preventing pollution at its source is central to combating AMR [18]. Likewise, the Quadripartite (WHO/FAO/OIE/UNEP) One Health Joint Plan of Action (2022–2026) includes AMR and specifically mandates integrating environmental considerations (action tracks 5 and 6) into AMR efforts [19]. These frameworks emphasize that robust surveillance and pollution control are necessary across all sectors, despite many national plans still being under-resourced in their environmental components.

Few countries enforce strict limits on antibiotic emissions from factories, agriculture, or hospitals. The AMR policies in most areas focus on judicious use of antimicrobials in human and animal health; however, they provide minimal advice on how to control emissions to the environment. This exclusion leaves untreated or poorly treated waste streams as potential routes of entry for resistant organisms and genes into neighbouring ecosystems. Researchers claim that this needs to be changed: 'One Health' model needs to explicitly place environmental controls on equal footing with the human and animal intervention. In practical terms, this would involve establishing explicit standards of discharge levels of antibiotics in industrial and municipal waste, investing in emerging technologies in wastewater and manure treatment that eliminate drug residues, as well as monitoring the presence of antibiotics and AMR genes in soils and water. The practical steps can support these changes. Hospitals and health systems would be able to purchase pharmaceuticals from suppliers whose production processes are confirmed to be low-emitting, thereby providing market incentives for clean production. Antibiotic manufacturers may be regulated by regulatory bodies to treat or pre-treat effluent to ensure it contains the safe limit of discharge to the environment. Some effective measures that farmers can take to reduce antibiotic runoff significantly include reducing non-therapeutic uses of antibiotics, implementing effective manure management practices, and optimizing crop rotation and sequencing. On a larger scale, multilateral trade regimes or state-corporate investment schemes, which offer access to markets or technical assistance on the condition of environmentally responsible manufacturing and disposal of antibiotics, are being evaluated and indicate steps to ensure that economic incentives and responsible stewardship are more closely aligned. These initiatives, along with others, align with the fundamental principles of One Health by formally establishing the integrity of the ecosystem as a quantifiable factor of human and animal health. At the same time, AMR surveillance should be combined with regular environmental surveillance of surface and groundwater, soil, and sediments to determine the presence of geographically and biologically distinct AMR deposits outside healthcare facilities. Delay is an expensive phenomenon, and every year of uncontrolled contamination adds reservoirs, albeit to some degree, selecting more and more resistant phenotypes and increasing the treatment burden.

Preserving antibiotic effectiveness thus demands a response from States, trade blocs, and civil society, which should address persistent inattention to environmental AMR sources. Only when we designate antibiotic-contaminated soils, waters, and aerosols as security and health priorities, subject to the same scrutiny and funding as stewardship of the pharmaceuticals, will the win over AMR be durable. No single constituency can outsource this challenge to another; our policies, clinical comportments, manufacturing, and consumption habits must mutually reinforce pollution minimization from every entry point. In short, protecting the efficacy of our remaining antibiotics now requires the systematic and accountable management of environmental contamination as an intrinsic dimension of global and planetary health.

This perspective synthesizes literature rather than providing primary source data. Some of the quantitative data cited are based on modelled or secondary data. They may vary geographically due to differences in the quality of reporting and the degree of environmental monitoring. While there is considerable evidence linking antibiotic pollution to the environmental selection of resistance genes, quantifying the specific environmental exposures that directly cause clinical AMR remains a challenge. Additionally, most widely cited studies lack coverage in low- and middle-income countries. More integrated environmental, clinical, and genomic surveillance is needed to accurately assess the global impact of pollution on antimicrobial resistance.

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Declarations

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